MIL-A-22523E(SHIPS)
15 May 1968
SUPERSEDING
MIL-A-22523D(SHIPS)
20 March 1967
(See 6.5)

MILITARY SPECIFICATION

ANALYZER-INDICATOR, DEAD RECKONING

(DRAI) (NAVAL SHIPBOARD)

1. SCOPE

1.1 Scope. This specification covers the design of Dead Reckoning Analyzer-Indicator (DRAI) for use in Naval ships. (DRAI will imply MK9 or later issue equipment when referred to in this specification.)

1.2 Classification. This specification covers two types of DRAI, as specified (see 6.2), each of which shall be identical in all respects except for the interface requirements and displays specified in 1.2.1 through 1.2.3.

1.2.1 Inputs. Both type I and type II DRAI shall accept heading and speed information in the following form:

OWN SHIP'S HEADING - 360 degrees per revolution and 10 degrees per revolution (1X and 36X), 115 volt, 400 Hz synchro information.

OWN SHIP'S SPEED - 100 knots per revolution and 10 knots per revolution (1X and 10X), 115 volt, 400 Hz synchro information.

1.2.2 Displays. Each type of DRAI shall compute and display the following information:

Type I	Type II
Own ship's heading.	Own ship's heading.
Own ship's speed (total).	Own ship's speed (total).
Own ship's distance traveled (total).	Own ship's speed (N-S).
Own ship's distance traveled (N-S).	Own ship's speed (E-W).
Own ship's distance traveled (E-W).	Latitude.
	Longitude.

1.2.3 Outputs. Each type of DRAI shall compute and transmit the following information:

Type I	Type II
Own ship's distance traveled (N-S)[1/]. Own ship's distance traveled (E-W)[1/]. 115 volt direct current (d.c.) step by step signal 750 turns per mile.	Own ship's distance traveled (N-S)[1/]. Own ship's distance traveled (E-W)[1/]. 115 volt d.c step by step signal, 750 turns per mile. Own ship's speed (N-S) 100 knots per revolution and 10 knots per revolution (1X and 10X), 115 volt, 400 Hz synchro signal. Own ship's speed (E-W) 100 knots per revolution and 10 knots per revolution (1X and 10X), 115 volt, 400 Hz synchro information. Latitude 360 degrees per revolution and 10 degrees per revolution (1X and 36X), 115 volt, 400 Hz synchro signal. Longitude 360 degrees per revolution and 10 degrees per revolution (1X and 36X), 115 volt, 400 Hz.

[1/] See footnote on next page.

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1/ The DRAI shall drive at least four MK 6 Mod 4 or Mod 4A Dead Reckoning Tracers(DRT's) connected in parallel across the DRAI. Each DRT shall contain two step motors per channel (N-S channel or E-W channel). The step motors are a highly inductive load, each step motor having a static impedance of approximately 1.5 henry per leg and a d.c resistance of approximately 700 ohms. Each DRAI step transmitter shall contain adequate suppression circuitry to avoid deleterious effects of transients generated by the DRT step motors on the step transmitter itself or on any other transient sensitive assembly within the DRAI. The suppression circuitry shall be designed for the worst case situation of six DRT's connected in parallel across the DRAI. There is no transient suppression circuitry of any kind in the DRT's.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-25	- Capacitors, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metallic Cases), General Specification For.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-I-983	- Interior Communication Equipment, Naval Shipboard; Basic Design Requirements For.
MIL-C-2194/2	- Cable, Electrical, Power, Type DGSA, Two Conductor, 1000 Volts, For Shipboard Use.
MIL-P-15137	- Provisioning Technical Documentation For Repair Parts For Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-M-17185	- Mounts, Resilient; General Specifications and Tests For (Shipboard Application).
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts. Preparation For Delivery Of.
MIL-S-20708	- Synchros, General Specification For.
MIL-P-21549	- Product Quality Program Requirements For Fleet Ballistic Missile Weapon System Contractors.
MIL-M-23313	- Maintainability Requirements For Shipboard and Shore Electronic Equipment and Systems.
MIL-M-24100	- Manuals, Orders and Other Technical Instructions, For Equipment and Systems.
MIL-C-24145/12	- Cable, Electrical, Special Purpose, For Shipboard Use, Type 3SWA.

STANDARDS

MILITARY

MIL-STD-470	- Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-471	- Maintainability Demonstration.
MIL-STD-701	- Lists of Standard Semiconductor Devices.
MIL-STD-721	- Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety.
MIL-STD-740	- Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781	- Reliability Tests and Exponential Distribution.
MIL-STD-785	- Requirements for Reliability Program (For Systems and Equipments).
MS17322	- Meter, Time Totalizing, Miniature, Digital, 115 Volt, 400 Cycle.

HANDBOOK

MILITARY

MIL-HDBK-217	- Reliability Stress and Failure Rate Data for Electronic Equipment.
MIL-HDBK-472	- Maintainability Prediction.

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PUBLICATIONS

NAVAL SHIP SYSTEMS COMMAND
NAVSHIPS 94324 -Maintainability Design Criteria Handbook for Design of Shipboard Electronic Equipment.
NAVSHIPS 94501 -Reliability Design Handbook.
NAVSHIPS 0900-002-3000 -Reliability and Maintainability Training Handbook.

NAVAL AIR SYSTEMS COMMAND
NAVWEPS 00-65-502 -Handbook Reliability Engineering.

U.S. NAVY OCEANOGRAPHIC OFFICE
Publication No. 9 -American Practical Navigator (1958 Edition).

(Application for copies should be submitted to the U.S. Navy Oceanographic Office, Washington, D. C. 20390 or the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606).

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)
239.01 - Measuring Methods and Practices Manual for Control of Spur, Helical and Herringbone Gears.

(Application for copies should be addressed to the American Gear Manufacturers Association, Standards Department, 1330 Massachusetts Avenue, N. W., Washington, D.C. 20005.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
Technical Progress Series
Volume IV - Manual for Reliability Control in Aerospace Equipment Development.

(Application for copies should be addressed to the Society of Automotive Engineers, Department M3, 485 Lexington Avenue, New York, N. Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to production, one preproduction sample of the equipment shall be tested as specified in 4.3 (see 6.3).

3.2 General features. The DRAI shall be in accordance with MIL-I-983, in addition to the requirements specified herein. Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.3 Description. The DRAI shall be a bulkhead mounted device which shall accept the signal specified in 1.2.1, display the information specified in 1.2.2 and provide the output signal specified in 1.2.3. All input and output signals shall be continuously received or transmitted. Own ship's heading and own ship's speed synchro information shall be received on 15CT4 synchros. All synchro transmitters shall be size 18. The DRAI shall operate within the accuracies specified in all ranges of latitude from 85 degrees North (N) to 85 degrees South (S). The DRAI shall include means to electrically slew the latitude and longitude counters and all associated computation circuitry to any desired value (see 3.7.6.1).

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3.4 Operation-General. The operation of the DRAI shall be as follows regardless of the type of DRAI specified: Own ship's speed information shall be received by a speed module which shall convert this information to an alternating current (a.c.) scaled voltage signal. This module shall also provide a readout of own ship's total speed. Own ship's heading information shall be received by a heading module which shall resolve the speed a.c. scaled voltage into N-S and E-W a.c. scaled voltage speed components. This module shall also provide a readout of own ship's heading (see 3.7.13). These component speed signals shall be integrated to obtain distance traveled component signals. Each of these distance traveled component signals shall be supplied to a step transmitter module for transmission as specified in 1.2.3.

3.4.1 Type I DRAI. In addition to the requirements specified in 3.4, the type I DRAI shall operate as follows: The speed module shall provide an output which shall be supplied to a total distance module where this signal shall be integrated and displayed as own ship's total distance traveled. The output of each integrator shall be supplied to component distance modules where these signals shall be integrated to obtain and display own ship's distance N-S and own ship's distance E-W.

3.4.2 Type II DRAI. In addition to the requirements specified in 3.4, the type II DRAI shall operate as follows: The speed component a.c. scaled voltage signals shall be supplied to component speed modules where own ship's speed N-S and own ship's speed E-W shall be displayed and transmitted as specified in 1.2.3. The output of the integrators shall be supplied to a latitude and a longitude module where latitude and longitude shall be displayed and transmitted as specified in 1.2.3.

3.5 Design. The DRAI shall be so designed that vibration, shock, or acceleration, which may be encountered in service onboard any of the types of ships for which the equipment is intended, will not adversely affect the accuracy of the equipment under normal conditions, nor, under extreme conditions, derange any part of the equipment beyond normal adjustment at sea. The equipment shall be so designed that the operator will not receive an electric shock when making any ordinary adjustment while the equipment is in operation.

3.5.1 Excessive play between moving parts, resulting in noise when the equipment is subjected to vibration, shall be eliminated insofar as practicable.

3.5.2 The equipment and its operation shall not be adversely affected by magnetic fields up to 5 gausses.

3.5.3 Adjusting devices, as well as the controls listed in 3.6.4, shall be free from excessive friction. Each adjusting device requiring a setting shall be provided with a suitable graduated dial, scale, or indicator, which shall be so designed that it may be read without difficulty. Handles of adjusting devices shall be so constructed or protected that accidental contacts will not change their setting, or shall be provided with protective covers attached to the unit in such a manner as not to cause interference or affect ease of handling. All components or modules requiring adjustment after repair or for calibration of the equipment as a whole shall be capable of being adjusted in place from the front without removal of any components in the equipment.

3.5.4 Mechanical components shall be of a suitable material, properly and evenly hardened or treated to prevent wear. They shall be provided with a means of adequate lubrication.

3.5.4.1 Mechanical resolvers, friction disks, rollers or other similar mechanical components shall not be used in this equipment.

3.5.5 Bearings shall be double-shielded, shall be at least class ABEC3 and shall be the prelubricated sealed-for-life type.

3.5.6 Windows, where required, shall be so mounted as to minimize danger of breakage or cracking due to stress at the edges.

3.5.7 All external wiring shall terminate at terminal boards and shall not be connected directly to internal switches, components, and so forth.

3.5.8 Parts of the equipment requiring accurate placing shall be marked or doweled, so that when disassembled for repairs or adjustments, the parts may be replaced in their proper position. Parts requiring occasional renewal due to unavoidable service wear shall be designed for maximum practicable ease of replacement.

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3.5.9 Interchangeability. Except as specifically provided herein, similar parts and permanent assemblies, including repair parts of corresponding apparatus furnished on the same contract or order or built to the same drawings shall be interchangeable without the necessity of further machining or hand fitting of any kind.

3.5.10 Proposed changes in design affecting interchangeability of parts or assemblies, including repair parts, shall be satisfactory to the Naval Ship Engineering Center (NAVSEC) (see 6.4) before being made effective.

3.5.11 Terminal boards shall be provided with protective covers to prevent accidental short circuits while working on the analyzer.

3.5.12 The effect of roll and pitch shall not adversely affect the operation of any unit of equipment when tested in accordance with 4.6.4.

3.5.13 Mercury and radium in any form shall not be used in this equipment or in its manufacture or testing.

3.5.14 Gears. Gears used in critical backlash applications shall be precision 1C or better as defined in Publication AGMA 239.01 and of a material satisfactory to NAVSEC (see 6.4). Corrosion resisting steel gears shall be used where this material will provide superior performance. Low torque, high precision gears shall not be lubricated unless they are fully sealed in an enclosure.

3.5.15 Warm-up time. The equipment shall be capable of performing all required functions within the specified accuracy limits (see 4.6.1) two minutes after power application.

3.5.16 Special tools. Special tools required for this equipment shall be submitted for approval to NAVSEC (see 6.4) and provided as part of the equipment. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.5.17 Provision shall be made to protect the DRAI against possible damage in case the equipment is allowed to run to or beyond N or S latitude limits, or at speed inputs in excess of 40 knots. Mechanical limit stops shall be provided on the latitude gear train at both the 85 degree N and 85 degree S positions, and on the total speed and the component speed gear trains at both the zero-knot and the 40-knot positions to prevent exceeding the operating range of the DRAI due to malfunction of equipment.

3.5.18 Integrators. Pulse-stepper motor integrators or equivalent shall be used wherever integration is performed.

3.6 Mechanical features. -

3.6.1 Enclosure. All parts required for the proper operation of the DRAI shall be mounted in an enclosing case (degree of enclosure dripproof or better) designed for bulkhead mounting. Provision shall be made for the protection of gears from dust. For ease of installation, terminal blocks shall be located at a minimum distance from the external wiring entry point with no components preventing direct access of any connector to the front of the appropriate terminal. Mounting and external wiring of the equipment shall be accomplished without removal of components from the enclosure and without exposing the gears to dust. Internal subassemblies shall not be supported by the enclosure. Adequate natural ventilation shall be provided to permit proper operation at any ambient temperature within the limits specified in MIL-I-983.

3.6.2 Accessibility. All units, subassemblies and modules shall be easily accessible for repair and adjustment. Provision shall be made for protection of all gears and gear trains from dust when the DRAI enclosure is in the opened position. All modules and mechanical subassemblies shall be held in place using captive fasteners only. All electrical connections between modules and subassemblies shall be made via plug-in type connectors. Under no circumstance shall it be necessary to unsolder any wire or disconnect any wire from a terminal board to remove and replace any module, subassembly or unit established by NAVSEC (see 6.4) as replaceable or repairable onboard ship, except for standard replacement parts such as synchros, resolvers, motors, etc. (see 3.6.2.2). Knurled knob captive screws with a screwdriver slot shall be used on all covers that require removal or opening for operating adjustment. All electronic circuits shall be mounted on printed circuit cards and located in a card file. These printed circuits shall

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be of a plug-in configuration and shall contain test points at the protruding end to permit troubleshooting when the cards are installed in the card file. The circuitry contained on each card shall correspond to a functional block of the system or module of which it is a part and shall be represented as such in the equipment technical manual (MIL-M-24100).

3.6.2.1 Breakout cables. Breakout cables shall be provided with each equipment for observing any module removed from the equipment. The cables shall contain a test panel for measuring input and output signals or monitoring critical points within the module during maintenance actions such as zeroing synchros, aligning the resolvers, etc. The breakout cable shall be six feet in length with the test panel being two feet from the end connected to the module. Suitable supports for each module shall be furnished with each equipment to allow repair work on the module without the unit resting on electrical components, connections and connectors or on components that may be disarranged or otherwise damaged. Printed circuit extender cards shall be provided for taking measurements within any printed circuit card in the equipment while the equipment is in operation. These extender cards shall plug into the slot occupied by the circuit card in question and shall have a connector on the opposite end into which the printed circuit card will fit. There shall be an electrical connection between opposing terminals at each end of the extender card. Provision shall be made for stowage of all breakout cables, module supports and extender cards within the equipment enclosure. Stowage shall be designed such that vibration and shock shall not result in audible vibrations of these parts or damage to either the parts or the equipment. All environmental testing to be performed on the preproduction unit shall be carried out with the breakout cables, extender cards and module supports within the equipment.

3.6.2.2 Maintenance philosophy. The following definitions of equipment levels shall be used for establishment of maintenance philosophy:

(a) EQUIPMENT - Equipment shall refer to the entire DRAI, all input and output terminals, casting and shockmounts.

(b) ASSEMBLY/MODULE - The assembly/module shall be the highest level functional block beneath the equipment level. Examples shall be own ship's speed module, own ship's heading module, latitude module, etc.

(c) SUBASSEMBLY - Subassemblies shall refer to those levels of equipment function which comprise a module. Usually electronic groupings such as printed circuit cards shall be referred to as subassemblies.

(d) COMPONENTS - Components shall refer to devices which perform a well defined unique function and which usually may not be broken down into a lower level. Synchros, resolvers, transformers, buffer amplifier, etc. shall be classified as components.

(e) PARTS - Parts shall be defined as the lowest level of equipment functional level. Parts are devices which by themselves do not perform a defined function in the equipment but rely on their relation with other parts or subassemblies. Resistors, capacitors, gears, bearings, etc. are defined as parts.

Maintenance shall be planned for this equipment on the ASSEMBLY/MODULE level; that is, the modules and assemblies shall not be replaced but shall be repaired onboard ship using necessary subassemblies and components to carry out necessary repairs. Shipboard maintenance shall include necessary calibration and alignment of components and subassemblies within the equipment. Subassemblies and components shall be replaced but not repaired onboard ship. The equipment technical manual shall contain complete repair and troubleshooting information for all levels of the equipment.

3.6.3 Weight and size. The weight of the DRAI shall not exceed 225 pounds. The size shall not exceed any of the following dimensions including vibration-shock isolators if required:

(a) Width - 23-3/4 inches.
(b) Height - 21 inches.
(c) Depth - 18-1/2 inches.

Size restrictions for submarine installation as required in MIL-I-983 applies. The DRAI shall mount to the bulkhead by means of twelve 1/4-inch diameter screws in the hole pattern shown on figure 1. Center of gravity location shall not deviate from the geometric center of the DRAI case by more than 10 percent of the overall dimension of each side when that side is taken as a reference plane.

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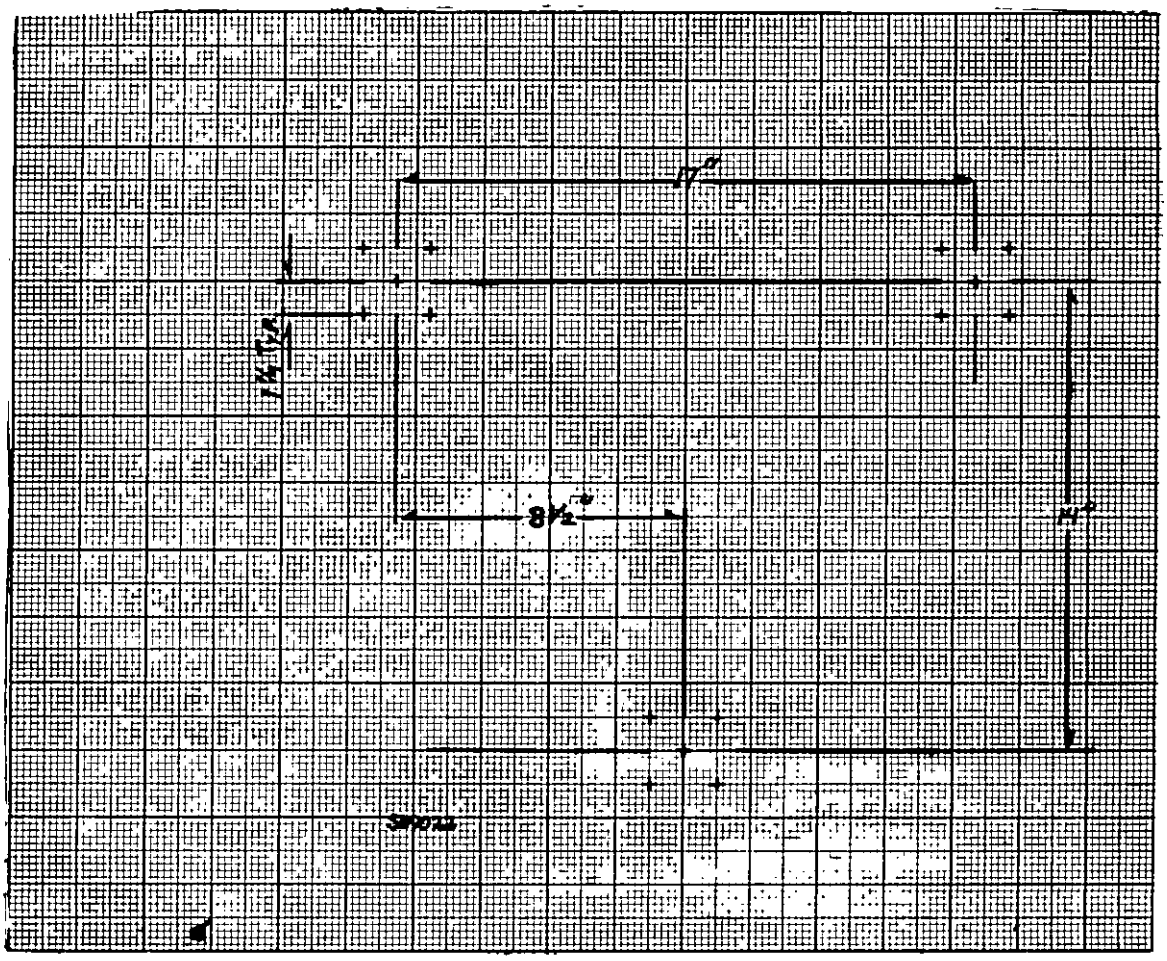


Figure 1 - Hole pattern for screws.

3.6.4 Controls and indicators. The following operating controls and indicators shall be provided on the cover of the enclosure (see 3.5.3):

(a) Latitude, longitude, slew switch controls (see 3.7.6.1).
(b) Blown fuse indicators (see 3.7.5).
(c) Heading indicator dial (see 3.7.13).
(d) Latitude and longitude, distance and velocity counters (see 3.7.6).
(e) Illumination control (see 3.7.14).
(f) On-off power switch and power-on indicator lights (see 3.7.16 and MIL-I-983).
(g) Distance reset control (see 3.7.6).
(h) Mode control (see 3.7.18).
(i) Integrator test jacks: Own ship's distance,
Latitude and own ship's distance north,
Longitude and own ship's distance east.
(j) Elapsed operational time indicator (see 3.7.17).

3.6.5 Shockmounts and isolators. The DRAI shall be designed to accommodate external resilient shockmounts in order to meet the requirements of MIL-I-983. These shockmounts shall be three in number arranged so that two are above the center of gravity and one below and are of a type approved by NAVSEC (see 6.4). When internal vibration or sound isolators are required in order to meet 4.6.2 and MIL-I-983, the isolators so used shall not permit the isolated structures to make contact with fixed structure under any conditions of attitude or externally applied shock. All shockmounts and isolators shall be provided as part of the DRAI and shall be in accordance with the requirements of MIL-M-17185.

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3.7 Electrical features.

3.7.1 Servo amplifiers. The servo amplifiers shall operate on 400-cycle, 115-volt synchro signals, shall be transistorized, and shall consist of the necessary items arranged on a removable chassis. The excitation lines for the synchro transmitters shall be appropriately fused within the DRAI. The total power taken from this source shall not exceed 150 watts. The amplifier chassis shall be grounded to the case, and a B minus bus isolated from the chassis shall be provided for the servo amplifiers, which shall be positively connected (bonded) to the case of the DRAI at one physical point only. Matching transformers shall be used between amplifier output and servo motor control field if the servo motor is not directly matched to the amplifier during manufacture. The servo motor shall not be modified to match the impedance of the amplifier. All servo trimming circuit shall be external to the amplifier package. Servo amplifiers shall have adjustable gain controls capable of adjusting the gain over a plus or minus 6 decibel (db) range.

3.7.2 Synchro unit. The synchro unit shall consist of the input synchro control transformers, a two-phase servo motor, gear trains, and the output synchro signal generators to provide for the receipt of the input synchro signal from the master gyrocompass, and log, and the retransmission of the computed and converted synchro signals to control type synchro signal receivers as required. The plate supporting the synchro control transformers, a servo motor, and synchro signal generators shall be so arranged as to be removable in order to adjust the synchro units for electrical zero. The synchro generators and control transformers shall be contractor furnished material in accordance with MIL-S-20708. The synchro equipment shall have corrosion-resistant steel housings, terminal board type lead terminations, spline shafts and shall be high temperature approved (125°C.±2°). The DRAI shall be protected against damage or permanent misalignment due to an incorrectly phased or excessive input signals. Such protection shall take the form of positive servo stops or other means suitable to the design configuration.

3.7.3 Servo motors. The servo motors shall be a two-phase low-inertia design with grease sealed ball bearings of the type specified in 3.5.5. Continuously operating motors shall be permanently lubricated for long life, shall not generate more than 15 watts of heat for each motor, and shall not generate mechanical or electrical noise at a level which would prevent meeting the low noise requirements of 4.6.2.

3.7.3.1 Stepper motors. Stepper motors used in the individual integrator channels shall be either of the variable reluctance or permanent magnet type and of low inertia design with grease sealed ball bearings of the type specified in 3.5.5. Excitation for the motor shall be generated within the DRAI.

3.7.4 Power supply. Single phase, 400-cycle, 115-volt a.c. shall be provided to the DRAI for operation of the equipment. 115-volt d.c. shall be provided for step transmission power.

3.7.4.1 Input signal isolation. The excitation power for the speed and the course inputs shall be isolated from each other to allow operation of the DRAI when speed and course inputs are on different phases of 3 phase power or are provided from different sources of 400 cycle power.

3.7.5 Blown fuse indicators. Blown fuse indicators shall be located on the front of the DRAI to indicate loss of power supply and loss of excitation to each of the separate circuits.

3.7.6 Latitude and longitude distance and velocity counters. The latitude and longitude counters shall read in degrees and minutes and shall indicate .2 minutes. They shall be equipped with an indicator to show correct hemisphere by the letters N, S, E, W as appropriate and reversing devices so that the latitude and longitude transmissions reverse when the ship crosses the equator, the prime meridian, and the 180th meridian. The distance counters shall read in miles and tenths of miles and indicate up to 9999.9 and shall be capable of being reset to zero from the inside of the case. The N-S and E-W distance counters shall increase in the number of miles indicated for north and east movement of the ship and shall decrease for movements to the south and west. The velocity counters shall read in .1 knots.

3.7.6.1 Slew switches. Two electrical switches shall be provided for resetting of the latitude and longitude counters in either direction at either fast or slow slew rates. The fast slew rate shall cause a counter-rate of change of at least 10 degrees per minute. The slow slew rate shall cause a counter slew rate of not more than 2.00 degrees per minute.

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3.7.7 Synchro capacitors. The correct delta connected synchro capacitors shall be provided for all synchro control transformers and synchro control differential transmitters receiving synchro transmissions from external sources to maintain unity power factor. These capacitors shall be mounted as close to the control transformers and synchro control differential transmitters as possible. The capacitor section shall be rated for 600 volts d.c. or greater, and shall be in accordance with MIL-C-25. For 400-cycle application, capacitors shall be rated for 1000 working d.c. volts. Each synchro capacitor shall have the capacitance of its three capacitor sections balanced within 1 percent, as measured by the divergence of the capacitance of any one leg from the arithmetical mean of the three.

3.7.8 Phase shift. The DRAI shall operate within the accuracy requirements of 3.7.11 for all phase shifts between 15-degree lag and 5 degree lead caused by external synchro units in the synchro transmission system.

3.7.9 Error calculations. The distance input to the equipment is defined as the theoretical length of run in nautical miles, and is represented by "D". The distance input to a DRAI is determined by the output of the speed input transmitter. The distance output of a DRAI is determined by its distance N-S, E-W counter readings, or special testing device connected to the outputs of the analyzer.

3.7.9.1 Resultant error. The resultant error is the error in the resultant of the N-S, E-W component miles of the DRAI output, and is expressed by the following formula:

$$E_r = \frac{\sqrt{E^2 + N^2}}{D} \times 100$$

Where:

E_r = Resultant error, in percent.
D = Length of run in nautical miles, determined from the distance input to the unit.
E = The difference in the reading of the east component mileage counter or analyzer output indicator and the E-W component of the distance input.
N = The difference in the reading of the north component mileage counter or analyzer output indicator and the N-S component of the distance input.

3.7.9.2 Position error. The position error is the error in final position indicated by the latitude and longitude dials, measured from the calculated final position as computed by the Mercator sailing method (Bowditch, "American Practical Navigator" H. O. Publication No. 9, 1958 edition) and expressed in percent of the total theoretical distance input to the unit. (The method of computing the position error is shown in the Appendix to this specification.)

3.7.9.3 Speed component error. The speed component error is the error in resultant of the N-S, E-W components of own ship's speed as transmitted as outputs of the DRAI, when compared with the total speed input to the DRAI. It is expressed by the following formula:

$$E_S = \frac{S_T - \sqrt{S_{Eout}^2 + S_{Nout}^2}}{S_T} \times 100$$

E_S = Error in speed components.
S_T = Speed input to DRAI.
S_{Eout} = DRAI output of E-W component of ship's speed.
S_{Nout} = DRAI output of N-S component of ship's speed.

3.7.9.4 Integrator error. The integrator error is the error in the computation of distance in an individual channel (that is, N-S distance, E-W distance, own ship's distance or longitude) based on a perfect speed input inserted directly at the input to the integrator. (This test shall be performed at an initial latitude of 0 degrees latitude when checking the longitude channel.) It is expressed by the following formula:

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$$E_I = \frac{100 \times E}{D}$$

Where:

E_I = Integrator error in percent.

D = Length of run in nautical miles, determined from the distance input directly to the integrator.

E = The difference in the observed resultant distance and the theoretical distance input.

The integrator shall be linear, this is, the integrator error (percentage deviation of actual calculation from the theoretical answer) shall be linear over the entire range of speeds from 5 to 60 knots. Tests shall be run using speed increments every 5 knots from 5 to 60 knots for varying times to obtain equal runs of 100 miles. The nautical mile error shall be constant over these ranges in speed.

3.7.10 Synchronization. The DRAI, within capability of the own ship's course input signal, shall be self-synchronous from all displacements (including the 180-degree displacement) out of synchronism with the input synchro signal.

3.7.11 Accuracies. The error of the DRAI shall not exceed the following values under all test conditions specified in 4.6. (Calculations of position error are given in the Appendix to this specification.)

(a) Resultant error, percent: 0.50.
(b) Position error, percent: 0.75 percent from 0 degrees to 70 degrees latitude, 1.00 percent from 70 degrees to 85 degrees latitude.
(c) Speed component error:
0-2 knots: 2/10 of 1 knot.
2-5 knots: Percent: 2.0.
5-40 knots: Percent: 0.50.
(d) Integrator error:
When referred to the input the component output shall be within 0.50 percent of the theoretically computed terminal point. The above integrator component error is based on a perfect input of ship's speed as is defined in 3.7.9.4. This accuracy shall be maintained over simultaneous variations of line voltage, line frequency, and all environmental conditions.
(e) Total distance error, Percent: 0.50.
(f) Accuracy speed counters: The speed counters shall indicate the speed signals transmitted or received to within 0.05 knots, (see 3.7.6).

Note: All accuracies shall be determined by comparing the final form of the output of the DRAI with the theoretical value calculated from the signal inputs. The final form of the DRAI output shall be that present at the terminal board output for the signal in question and not the readout. If an output is to be displayed but not transmitted the counter readout shall be the DRAI output. The input to the DRAI shall be the signal presented at the input terminal boards. The value of the input shall be determined by using a dummy dual speed input signal for a reference.

3.7.12 Effect of synchro bus load on synchro control transformers. In order to prevent excessive reflected errors on the synchro bus, the secondary load on the synchro control transformers shall be greater than 30,000 ohms for 400-cycle units and greater than 30,000 ohms for 60 cycle units.

3.7.13 Heading indicator dial. A double dial with concentric faces shall be provided. The outer scale shall be calibrated over its entire periphery from 0 to 360 degrees in 1-degree steps with every 5-degree mark accentuated, and every 10-degree mark numbered to indicate the heading at 1 speed. The inner dial calibration shall read from 0 to 10 degrees in 0.1-degree steps with every 0.5-degree mark accentuated, and every 1-degree mark numbered. A single dial index shall be used which shall permit movement of approximately 3 degrees in either direction for adjustment purposes. This dial shall be capable of indicating changes of course at a rate of not less than 10 degrees per second.

3.7.14 Illumination control. All dials and counters shall be illuminated in red, and shall be controlled by a single rheostat or other control device, as specified in the applicable paragraphs of MIL-I-983.

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3.7.14.1 Red illumination. The illumination shall be red indirect light, free from glare and shadows and of uniform intensity throughout. No stray light shall be visible externally. Red illumination shall be arranged to utilize filters passing less than 1 percent of light of wave length shorter than 5800 angstrom units but passing at least 90 percent of light of wave length longer than 6800 angstrom units. Illumination of individual instruments shall be satisfactory to NAVSEC (see 6.4).

3.7.15 Step transmission system. Each step transmission in the DRAI shall be capable of continuously driving at least eight-step motors of a type normally provided in DRT's or other plotting equipment. Both positive and negative leads of the 115 volt d.c. shall be provided in the DRAI to allow reversing of leads, as necessary, for compatibility with DRT's and plotters. Arc suppression shall be part of the DRAI.

3.7.16 Power switch. The power switch shall break both sides of all power circuits and shall secure all forms and types of power within the unit including all information and excitation leads of all synchros in the input circuit (R1, R2, R3, S1, S2 of each synchro in the OSS and OSH modules) and illumination. A power-on indicator light shall be provided for both the 400 cycle power and the d.c. power inputs and shall be actuated by the power switch.

3.7.17 Time totalizing meter. A time totalizing meter shall be mounted in clear view on the front panel to provide a means of telling when the DRAI is to be calibrated. A meter similar to that described in MS17322 shall be used.

3.7.18 Calibration. Calibration shall be accomplished without using external test equipment or without the requirement for externally generated signals. The only tool required to calibrate the equipment shall be a screwdriver. A switch shall be provided which shall permit calibration of own ship's total distance (type I), own ship's distance North (type I and type II) and latitude (type II), own ship's distance East (type I and type II) and longitude (type II). A screwdriver adjustment shall be provided at each switch position to permit calibration. Calibration shall be capable of being accomplished without requiring any electrical circuit switching external to the DRAI. An elapsed operational time indicator shall be mounted in clear view on the front panel to provide a means of telling when the DRAI is to be calibrated (see 3.7.17). The DRAI shall be equipped with an internal calibration system which will permit rapid accurate calibration of the DRAI without the removal of any module, subassembly or component from its normal operating position and without disconnecting or changing any of the DRAI wiring. A mode control switch shall be provided with positions of operate and calibrate. In the operate or normal mode, the DRAI shall function from the own ship's speed and own ship's course input signals. In the calibrate mode each integrator shall be capable of a calibration check and adjustment. All test signals shall be generated within the DRAI. An integrator selector shall be provided to indicate which integrator channel is under check. A simple means (example: screwdriver) shall be the only equipment required to accomplish the calibration adjustments without removal of any module, subassembly or component from its normal operating position and without disconnecting or changing any of the DRAI wiring. A readout shall be provided within the DRAI to determine if the DRAI is in calibration, and if not in calibration, by how much it is out. The complete calibration procedure shall not require more than 10 minutes. Calibration procedures shall be furnished and affixed to the DRAI.

3.7.19 Electron tubes. No electron tubes, or thermionic devices of any type, shall be used in this equipment.

3.7.20 Semiconductor devices. Semiconductor devices shall be selected from MIL-STD-701. The contractor shall submit a tentative semiconductor complement report to NAVSEC (see 6.4) as soon as the equipment design requirements have been determined. In the event a change in the semiconductor complement is found necessary, it shall be reported within 30 days. In order to assure sufficient quantity for maintenance of equipment in the Naval service, the contractor shall furnish NAVSEC (see 6.4) with a final semiconductor complement report after the equipment design has been completed, and before production has begun. These reports shall be submitted on the same Form DD 816 as used for electron tubes, and shall be combined into a consolidated form for each equipment.

3.7.21 Synchro transmitted signals. Own ship's speed E-W, N-S, 10 and 100 knots per revolution; own ship's distance E-W, N-S, 1 and 360 revolutions per mile and latitude-longitude 10 and 360 degrees per revolution shall be transmitted by an 18TR4b synchro in accordance with MIL-S-20708.

3.7.22 Protection from external transients. All DRAI output circuits shall be capable of withstanding repeated transient impulses from external sources of 800 volt magnitude and having a time duration of 1 millisecond.

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3.7.23 Protective coatings. Use of any impregnating, embedding, incapsulating, conformal or protective coating of any kind shall be approved by NAVSEC (see 6.4) prior to its application to any electrical component, subassembly, assembly or module.

3.7.24 Terminal marking. All terminals (component and otherwise), terminal boards, plugs, jacks and connectors shall be marked in a clear and permanent manner so as to identify individual wires, terminals and other electrical connections. The identification designations used shall appear on all applicable drawings and shall be approved by NAVSEC (see 6.4) prior to its adoption by the contractor.

3.7.25 Terminal arrangements. The main terminals for connecting the DRAI wiring to that of the ship shall be located on a structural member with direct access to the cable entry point. The interunit wiring shall be on blocks located for convenience during maintenance (see 3.5.11). A minimum of 25 percent spare terminals shall be provided for possible future additions and modifications. All cables shall be brought out via stuffing tubes. A plate opening shall be provided for mounting stuffing tubes. Stuffing tubes shall be provided by the installing activity. All interconnection cables between the DRAI and external equipment shall be type 3SWA in accordance with MIL-C-24145/12 except for the 115 volt, 400 cycle and the 115 volt, d.c. power inputs which shall be type DSGA-4 cable in accordance with MIL-C-2194/2.

3.7.26 Isolation. Electrical isolation shall be provided within the DRAI isolating each synchro input signal, the a.c. power input, and each synchro output signal.

3.8 Fault isolation and equipment calibration. Means shall be provided on the front panel (see 3.6.4(i)) to calibrate the integrators and to isolate operational problems in the case of equipment malfunction (see 3.7.18). Source of DRAI malfunction shall be capable of being located without the removal of any module, subassembly or component from the DRAI enclosure. Source of DRAI malfunction is hereby defined as the lowest level of replaceable assembly which will correct the malfunction. Time to locate source of malfunction, repair and return DRAI to service in a calibrated condition shall not exceed 4 hours. The 4 hour time period does not include time to obtain replacement components. Approval by NAVSEC (see 6.4) shall be obtained prior to the incorporation of any module or subassembly in the DRAI whose electrical malfunction cannot be located with either a volt-ohm meter or a vacuum tube voltmeter.

3.9 The supplier shall furnish accurate information on weight and vertical center-of-gravity (v.c.g.) of each unit of equipment weighing 100 pounds or more, as well as items known to be stowed or group-located in quantity so as to constitute an aggregate weight of 100 pounds or more. This information shall be required as follows:

(a) By notation of calculated weight and location of v.c.g on the outline drawing initially submitted for approval.
(b) On all revisions of drawings which result from changes in the calculations under (a) above.
(c) By notation of the actually measured weight and corrected location of v.c.g., made prior to shipment, on the final revision of drawings sent to the Government.

3.10 Reliability/maintainability (R/M). The equipment shall provide satisfactory performance and operation in accordance with this specification for a 20 year period during which time it shall operate 6000 hours per year while experiencing 200 on-off cycles per year, evenly distributed throughout the year. The equipment shall operate in three month deployment (mission) periods, during which time the equipment shall accumulate a total operating time of 2000 hours, shall experience 100 on-off cycles, and shall require minimum preventive and corrective maintenance. After each three month/2000 hour cycle, limited prevention or corrective maintenance may be performed to insure satisfactory operation during the next three month/2000 cycle hour cycle. The equipment shall be designed and constructed such that after three years (18,000 hours) of operation the reliability will not have been degraded by more than 15 percent of the value predicted. During this period scheduled replacement of parts shall not be required to maintain the equipment reliability within this specification. After this three year (18,000 hour) period, a detailed overhaul procedure will be accomplished to restore the equipment MTBF to within 2 percent of the value originally predicted. Detailed overhaul procedures and part requirements shall be included in the "Plan for Maintenance" (see 3.10.11.3).

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3.10.1 During each three month/1000 hour deployment, the equipment shall have an operational availability (Ao) of 99.9 percent for a type I and 99.8 percent for a type II, defined as follows:

$$Ao = \frac{(2000 - \bar{M}t \times 100)}{2000} \%$$

where $\bar{M}t$ is the mean total maintenance downtime for a three month/2000 hour mission defined as:

$$\bar{M}t = \bar{N}c \times \bar{M}ct + \bar{N}p \times \bar{M}pt$$

where $\bar{N}c$ = mean number of corrective maintenance actions required during a 2000 hour period of operation ($\bar{N}c = \frac{2000 \text{ HOURS}}{\text{MTBF}}$).

$\bar{M}ct$ = mean corrective maintenance downtime for a corrective maintenance action. The maintenance downtime for corrective maintenance shall be predicted and demonstrated assuming that maintenance personnel and any necessary test equipment and repair parts are immediately available. $\bar{M}ct$ is the time required to analyze the equipment malfunction, localize the defective part or parts, execute the required maintenance actions and restore the equipment to a satisfactory operating condition.

$\bar{N}p$ = mean number of preventive maintenance actions required during a three month/2000 hour mission. The value of $\bar{N}p$ shall be demonstrated by accuracy testing each equipment undergoing the MIL-STD-781 reliability demonstration after each 100 hours of operation. The results of these tests shall be used to calculate a mean time between required calibration ($\bar{m}p$). $\bar{N}p$ shall be determined as $\frac{2000}{\bar{m}p}$.

$\bar{M}pt$ = mean preventive maintenance downtime. Replacement of parts or modules during the three month/2000 hour mission shall not be required as part of the preventive maintenance schedule.

3.10.2 The contractor shall predict, prior to formal demonstration, the values of each parameter affecting the specified availability. This prediction shall become binding and the contractor shall be committed to achieving each of the parameters listed. Should formal demonstration indicate that any parameter has not measured up to the prediction, the contractor shall conduct a design review and failure analysis to determine design deficiencies and shall make any necessary design changes to the equipment. The demonstration shall be repeated to verify that the prediction has been achieved. Upon completion of the reliability demonstrations all equipments tested shall be retrofitted prior to delivery at contractor's expense as required by the "Plan for Maintenance" and reliability analysis. Any design changes, including use of new parts or modification of manufacturing techniques shall be retrofitted in all equipments previously delivered, changed in all equipment to be manufactured and changed in all equipments not shipped from contractor's facility and shall be reflected in all documentation submitted to the Government as required by the DD 1423. All data submitted, or to be submitted, as required by the DD 1423 shall be changed at contractor's expense to reflect the engineering design changes.

3.10.3 The listing below indicates the reference which shall be used in predicting each of the parameters, the reference which shall be used to demonstrate the parameters, and the minimum acceptable values of each parameter. The minimum acceptable values listed are for use in performing R/M tradeoff studies and are not to be interpreted as compliance with operational availability requirements. The minimum acceptable MTBF requires a specified MTBF (predicted) not lower than 1.5 times the minimum acceptable value specified below (see MIL-STD-781).

PARAMETER	PREDICTION	DEMONSTRATION	MINIMUM ACCEPTABLE VALUES	
			Type I	Type II
MTBF	MIL-STD-756	MIL-STD-781, test level B, test plan II	1500 hrs.	1200 hrs.
$\bar{M}ct$	MIL-HDBK-472, method 2	MIL-STD-471, method 2	2.5 hrs.	2.5 hrs.

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PARAMETER	PREDICTION	DEMONSTRATION	MINIMUM ACCEPTABLE VALUES	
			Type I	Type II
$\overline{N}p$		MIL-STD-781 test level B, test plan II	2	2
$\overline{M}pt$	MIL-HDBK-472, method 2	MIL-STD-471 method 2	1 hr.	1 hr.

3.10.4 Reliability demonstration shall be conducted according to test level B, test plan II of MIL-STD-781. The specified MTBF for this demonstration shall be that predicted by the contractor; however, the minimum acceptable MTBF shall not be less than that specified. During each accuracy test after each 100 hours of operation, should the equipment fail to comply with the accuracy requirements of this specification the number of operating hours since the last calibration shall be noted and logged. The equipment shall be recalibrated and the test continued. Calibration of equipments during this period shall be part of the MIL-STD-471 demonstration. For demonstration of MTBF, a failure (as defined in MIL-STD-721) shall not be considered if correction can be accomplished be recalibration. Any failure which cannot be corrected by calibration shall be considered corrective maintenance and the failure logged per MIL-STD-781. All maintenance actions and demonstrations shall be carried out using only those parts permitted by the onboard maintenance philosophy established for this equipment (see 3.6.2.2). All maintenance actions will be conducted in accordance with the philosophy specified in 3.6.2.2.

3.10.5 The supplier shall provide and maintain a detailed written reliability (R) design and maintainability (M) design assurance program plan in accordance with the general requirements of MIL-STD-470, MIL-STD-721 and MIL-STD-785. The plan shall be acceptable to NAVSEC (see 6.4) and shall be submitted as a separate and complete entity within the supplier's proposal for designing and developing this equipment. The plan shall be an integrated effort within the total program plan and shall comply with the reliability program plan of MIL-STD-470 and MIL-STD-785 as a minimum.

3.10.6 The R design assurance plan shall include the following in accordance with MIL-STD-721 and MIL-STD-785:

(a) R organization.
(b) R management and control.
(c) R program review.
(d) Manner of demonstrating R.
(e) R design guides.
(f) Operation modes determination.
(g) Equipment apportionment prediction and math models.
(h) Failure mode and effects analysis (materials).
(i) Human error analysis (man-machine interface).
(j) Integrated R and M test planning of MIL-P-21549 and MIL-STD-785.
(k) Integrated R and M design review.
(l) Quality assurance planning during design.
(m) Supplier and subcontractor R and M programs.
(n) Failure data collection, analysis and corrective action.

3.10.7 The M design assurance plan shall include the following in accordance with MIL-STD-470:

(a) M organization.
(b) M management control.
(c) M program review.
(d) Manner of demonstrating M.
(e) M design guidelines.
(f) M engineering apportionment and R and M design trade of math models.
(g) M engineering prediction.
(h) M task and skill analysis.
(i) Integrated R and M test planning.
(j) Integrated R and M design reviews.
(k) Supplier and subcontractor M program control.
(l) M data collection, analysis and corrective action system.
(m) M retention (quality assurance plans) during production, installation and so forth.
(n) Logistic support data, including maintenance manuals and repair parts.

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3.10.8 The R design assurance program shall incorporate the following specific features:

(a) Operation modes determination - NAVWEPS 00-65-502, chapter 2 shall be used as a guide.
(b) R design guidelines - NAVSHIPS 94501, and MIL-HDBK-217, shall be a basis for design guideline criteria, as well as supplier internal design guidance documents.
(c) Apportionment and math models - NAVSHIPS 0900-002-3000, chapter 3 and chapter 6 shall be used respectively as guides.
(d) Failure mode and effect analysis - NAVSHIPS 0900-002-3000, chapter 12 and chapter 6 of SAE Technical Progress Series, Vol. IV.
(e) Human error analysis - NAVSHIPS 0900-002-3000, chapter 14 shall be used as a guide.
(f) Supplier and subcontractor control - NAVSHIPS 0900-002-3000, chapter 19 shall be used as the basic guide.
(g) Failure data collection, analysis, and corrective action - NAVSHIPS 0900-002-3000, chapter 16 and MIL-P-21549, shall establish the basic minimum criteria for inclusion in the R program.

3.10.9 The M design assurance program shall incorporate the following specific features:

(a) Maintenance concept - NAVSHIPS 0900-002-3000, chapter 8 shall be used as a guide. Definition shall comply with MIL-STD-721.
(b) Manner of demonstrating M - NAVSHIPS 0900-002-3000, chapter 8 and chapter 11 shall be used as guides.
(c) M design guidelines - NAVSHIPS 94324 shall be used, as well as contractor internal guides and check lists.
(d) R and M design tradeoffs apportionment and math models - NAVSHIPS 0900-002-3000, chapter 8, shall be used as a guide.
(e) M prediction - MIL-M-23313 shall be used.
(f) M task and skill analysis - NAVSHIPS 0900-002-3000, chapter 8, figures 8-37 and 8-38 shall be used as a guide.
(g) Supplier and subcontractor M control - NAVSHIPS 0900-002-3000, chapter 19 shall be used as a guide.
(h) M data collection, analysis and corrective action - NAVSHIPS 0900-002-3000, chapter 16 shall be used.
(i) Integrated R and M design reviews - NAVSHIPS 0900-002-3000, chapter 15 shall be used as a guide.
(j) M retention during production, installation, and so forth - The contractor shall integrate with quality assurance plan and identify problem areas and interfaces.
(k) *Logistic support data, including maintenance manuals and spare/repair* parts - NAVSHIPS 0900-002-3000, chapter 14 and chapter 20 shall be used as guides. Correlation between R program and maintenance support information and hardware shall be included by the contractor.

3.10.10 Reliability/maintainability program plan. This plan prescribes the reliability tasks which the contractor shall implement throughout the design, construction and test of the DRAI. The purpose of this program plan is to facilitate the design and production of equipments in minimum time at minimum cost and in a manner which gives maximum confidence that specified R/M parameters are being attained.

3.10.10.1 Redundance considerations. In the event the contractor chooses to use redundancy as a means of achieving the required reliability the following shall apply:

(a) Standby. Switching redundancy is preferred (i.e., no power applied to the redundant element until failure of the primary element).
(b) If active (power applied to redundant element continuously) redundancy is employed, it shall be the switching type only. No load sharing will be permitted.
(c) The switch-over from the failed element to the redundant element may be accomplished by either manual switching from the control panel or by automatic means. In either case, indicators must be provided to indicate which element is in use.

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3.10.10.2 Maintenance considerations. It is the intention of this specification to provide a system that will require minimum maintenance and calibration during a three month ship's deployment during which time the system will be operated approximately 2000 hours with 100 on-off cycles. At the end of the three month deployment period, the system will undergo a detailed maintenance/calibration schedule which assures a continuing operational availability for the next three months usage. It is desirable that the maintenance/calibration time be kept to an absolute minimum by providing self-contained test features and readily accessible and repairable modules.

3.10.10.3 Philosophy. There are five fundamental concepts which shall be employed in achieving the R/M objectives of this specification. The policies and procedures which result from these concepts will facilitate the timely and orderly accomplishment of R/M tasks.

3.10.10.3.1 Inherent equipment reliability and maintainability shall be achieved through concerted design effort early in the program.

3.10.10.3.2 Design reliabilities shall be translated into equipment reliabilities by carefully controlled parts procurement and production programs.

3.10.10.3.3 The objectives of developmental tests and formal reliability demonstrations are distinctly different and natually exclusive. Developmental tests are a check of design effectiveness and parts adequacy which assure the contractor of a quality product. Formal R/M demonstrations are statistically valid tests of hypothesis which give NAVSEC a known confidence that specified reliability parameters have been achieved.

3.10.10.3.4 A single contractor R/M manager shall be responsible and accountable for planning, implementing, controlling, and reporting all R/M tasks required by this specification.

3.10.10.3.5 The contractor shall require NAVSEC approvals at specified program points, and he shall be constrained from commencing tasks which follow sequentially whenever approvals are withheld for failure to comply with contractual requirements.

3.10.10.4 The tasks to be performed by the contractor are divided into four categories: (1) design related tasks, which are performed prior to equipment construction; (2) development tasks, which occur as the completed design is translated into equipment; (3) formal demonstrations, which verify that the equipments meet specified parameters; (4) report submissions, which occur throughout the program. The tasks are specified in 3.10.10.4.1 through 3.10.10.4.4, in the order in which they will occur during the procurement.

3.10.10.4.1 Design tasks. During the design phase of the program great emphasisall be placed on controlling the efforts of individual equipment design engineers. The R/M manager shall establish formal, detailed departmental procedures for accomplishing these tasks, and shall supervise the implementation of the procedures.

3.10.10.4.1.1 Design engineer indoctrination. The R/M manager shall thoroughly instruct each equipment design engineer in the techniques for achieving circuit reliability. Engineers shall understand the effects of environmental stresses on component life. They shall be made familiar with techniques of component derating and the use of failure-rate prediction documents such as MIL-HDBK-217. The necessity for employing standard circuits shall be emphasized, and the design engineers shall be made aware of sources of standard parts data and parts performance and reliability data. Design engineers shall be indoctrinated in the use of design review worksheets and they shall be briefed on the need for fully satisfying all circuit/component reliability design requirements. Emphasis shall be placed on the considerations which determine optimal module configurations. The importance of ready module accessibility, adequate provisions for fault isolation, and effective circuit and personnel safety devices shall be stressed. The responsibilities of the design engineers to the R/M manager shall be clearly established.

3.10.10.4.1.2 Initial apportionment of reliability requirements. The R/M manager shall prepare a model which will show the logical devisions of the system into assemblies, subassemblies, and modules, and he shall assign to each a quantitative reliability requirement based upon parts complexity and overall system reliability requirements. The assigned values shall be based upon state-of-the-art technology and shall reflect a liberal margin of design safety.

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3.10.10.4.1.3 Design engineer requirements. The R/M manager shall provide each design engineer with the apportioned failure rate for each system element he is to design. Each design engineer shall be given the parts derating policies for components, and he shall be furnished design review worksheets for each element he is to design. As each element design is finalized, the design review worksheets shall be completed and turned in to the R/M manager for review.

3.10.10.4.1.4 Informal design reviews. The R/M manager shall conduct an informal design review with each design engineer at least weekly. During these reviews, circuit features should be discussed, the derating of components scrutinized, unique maintainability features discussed, and the design review worksheets examined. The design engineer should discuss any problems he is encountering in meeting the apportioned failure rates, and if necessary the manager shall reapportion these rates among the circuits on modules.

3.10.10.4.1.5 Preliminary design review (PDR). As part of the PDR the R/M manager shall review the progress of the reliability design to date. The following items, as a minimum, shall be included in a technical report to NAVSEC one week prior to the PDR meeting:

(a) A block diagram showing the apportioned failure rates to the module level.
(b) A summary of informal design review activities.
(c) A summary of any problem areas uncovered to date.
(d) A summary of unique or significant reliability and maintainability features developed.

PDR shall be the first program point where NAVSEC approval is required of the contractor's R/M progress.

3.10.10.4.1.6 Reliability math model. The R/M manager shall prepare a reliability prediction model which reflects data taken from the design review worksheets. The prediction shall show a block diagram of elements in serial or redundant configuration as applicable. The model shall be prepared using a parts complexity model and MIL-HDBK-217 failure rates. Each block shall show the elements name and identification number, and the predicted element failure rate. Failure rates used other than those contained in MIL-HDBK-217 should be noted and their source listed. NAVSEC approval of non MIL-HDBK-217 failure rates is required by the contractor.

3.10.10.4.1.7 Failure mode and effect analysis. As an integral part of the early design phase, the R/M manager shall develop projected analyses for the system to determine possible modes of failure and their effect on mission success. The primary objective of these analyses shall be to discover critical failure areas and remove susceptibility to such failures from the system. The analysis shall be made starting at the system level and expanding downward to the component level. In the analysis, each potential failure should be considered in light of probability of occurrence and should be categorized as to probable effect on mission success.

3.10.10.4.1.8 Parts procurement. The R/M manager shall review all lists of parts to be purchased, before purchase requests are released to procurement. He shall insure that nonstandard or tested and nonapproved parts are not ordered unless acceptable parts are unavailable from any source.

3.10.10.4.1.9 Design approvals. As each design is completed, and when the resulting parts list and drawings have been reviewed by the R/M manager, the manager shall sign-off the design. Design sign-off for a module certifies that the apportioned failure rate has been met, and the specified parts list satisfies contractual requirements.

3.10.10.4.1.10 Critical design review (CDR). During the overall system CDR, the R/M manager shall summarize the status of the R/M design. He shall include in a technical report to NAVSEC one week prior to the CDR the following items as a minimum:

(a) A summary of the informal design reviews to date.
(b) A discussion of any reliability or maintainability problem areas.
(c) A list of signed-off designs, with dates.
(d) An updated reliability block diagram and math model.
(e) A list of all nonstandard or untested parts for which approval has been received or approval is pending NAVSEC action.
(f) Results of failure mode and effect analysis.

CDR is the second program point where NAVSEC approval of the contractor's reliability progress is required.

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3.10.10.4.2 Development tasks. Release of the contractor's design to production shall follow receipt of NAVSEC approval of the CDR. During the production phase, parts ordering is completed, production is initiated, equipment developmental tests are completed, and any required hardware or design changes are made.

3.10.10.4.2.1 Stress verification measurements. During the module factory test phase of the first production unit, actual measurements of component stress levels shall be made as a verification of stress level calculations. The purpose of these measurements is to insure that the design engineer's intended stresses have not been exceeded in each module. These measurements shall be made on 20 percent of all electrical components in each module of which 50 percent are active components (i.e., transistors, motors, etc.). The environmental temperature of each selected part, and its operating stress level shall be measured in the first module which is fabricated. The dates and results of all measurements shall be retained in a file.

3.10.10.4.2.2 Failure analysis and corrective action. During the developmental tests, formal failure reporting to NAVSEC will be performed for every failure that occurs. The R/M manager will review each failure report and ascertain that the cause of failure has been correctly determined. Additionally, he shall insure that adequate corrective action is taken and the fault is corrected.

3.10.10.4.2.3 Reliability/maintainability design review (RMDR). A formal review of the R/M program shall be conducted by the R/M manager following completion of developmental testing, and immediately prior to the start of formal reliability demonstrations. During this review the following information shall be reviewed:

(a) An updated reliability math model reflecting results of the stress verification measurements.
(b) A summary of completed corrective actions.
(c) A summary of failure reports.
(d) A list of all critical life items.
(e) A list of any outstanding untested or nonstandard parts awaiting approval.
(f) The results of all parts approved tests.
(g) The approved reliability demonstration test procedure.

The RMDR is the third program point where NAVSEC approval of the contractor's R/M progress is required.

3.10.10.4.3 Formal reliability demonstration. The predicted MTBF of the system shall be demonstrated by subjecting 21 equipments to an environmental life test under test level B of MIL-STD-781 and testing until an accept or reject decision is reached in accordance with test plan II of MIL-STD-781. Test plan II is designed for 20 percent decision risks and a 1.5 to 1 discrimination ratio: i.e., for demonstration of minimum acceptable MTBF it is designed such that there is a 20 percent chance that an equipment having a specified MTBF will be rejected (manufacturer's risk) and a 20 percent chance that an equipment having a minimum acceptable MTBF will be accepted (consumer's risk).

3.10.10.4.3.1 Formal maintainability demonstration. The contractor shall demonstrate the achieved degree of maintainability by testing the systems according to method 2 of MIL-STD-471. When possible, maintainability samples shall include equipments which failed during burn-in or during the reliability demonstration. Maintenance tasks shall be so varied such that at least 50 different tasks shall be sampled. Tasks shall be selected according to predicted usage rates from the reliability prediction.

3.10.10.4.4 Data submissions. The R/M manager shall prepare and submit the following data:

(a) R/M periodic progress reports (monthly).
(b) PDR and CDR technical reports (one week prior to formal meeting).
(c) Failure mode and effect analysis report (30 days after completion of analysis).
(d) Failure reports (10 days after a failure).
(e) Reliability test and evaluation report (60 days after completion of test).

3.10.10.5 Subcontractor reliability programs. The R/M manager shall review each subcontractor R/M specification to insure that quantitative requirements are passed on correctly.

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3.10.11 Maintenance studies and plans. The contractor shall conduct a Maintenance Engineering Analysis (MEA) utilizing the efforts of his reliability and maintenance design assurance plans to explore the possibility of tradeoff's between reliability and maintainability to develop the optimum system configuration.

3.10.11.1 The design shall be based on the fundamental precepts of simplicity, reliability, ruggedness and maintainability. The human element must be considered as early as possible and continually in system design in order to incorporate systems and equipments requiring a minimum of logistic and maintenance effort and to provide a ship system that can be operated and maintained efficiently and economically with minimal personnel requirements. Greatest dependability will result if the tasks required are straightforward thus minimizing the margin of error. The first level supervisor for shipboard and yard work shall be considered to be a high school graduate with special training.

3.10.11.1.1 The adequacy of logistic support is a matter equal in importance to the adequacy of the item itself and shall be so considered in design and production. The reliability and the maintenance and equipment overhaul program of the item throughout its 20 year life shall be tuned to assuring that the unit will operate satisfactorily under fire, during and after a mechanical shock similar to that specified in MIL-S-901. Any item subject to degradation through age, operation and environmental forces and which, therefore, could fail at time of the additional stress resulting from mechanical shock shall be considered in the reliability design assurance plans.

3.10.11.1.2 Under combat conditions, the only assured supply to the Fleet may be limited to the material covered by individual ships, the mobile logistic support forces and stocks positioned at overseas bases. The range and depth of Fleet support spare and repair parts and equipment related consumables to be carried by the forces afloat shall provide a 90 day endurance. The items to be carried by the mobile logistic support forces, and the resupply material positoned at designated bases in support of actual and planned fleet deployments without augmentation, shall satisfy the 90 day corrective and preventive maintenance requirements.

3.10.11.2 The contractor shall submit to NAVSEC (see 6.4) for approval, a "Maintenance Engineering Analysis Report" (MEAR) which will document and substantiate the MEA. As a minimum requirement, the MEAR shall include the information specified in 3.10.11.2.1 through 3.10.11.2.3.

3.10.11.2.1 A R/M design evaluation which, as a result of the MEA, shall establish the following:

(a) Military essentiality or critically of each part and subsystem.
(b) MTBF and reliability for each part and subsystem.
(c) Availability for a four day continuous mission.
(d) Modes of failure from analysis of physics of failure.
(e) Mean time to restore.

3.10.11.2.2 A discussion of possible tradeoffs indicating alternatives which will increase system reliability or reduce system down time.

3.10.11.2.3 Justification for tradeoffs which were made or not made.

3.10.11.3 Upon approval of the MEAR, the contractor shall submit to NAVSEC (see 6.4) for approval, a "Plan for Maintenance". The "Plan for Maintenance" shall be the framework around which all aspects of integrated logistic support shall be constructed. TThis "Plan" shall provide the technical data base for all logistic requirements. As a minimum requirement, the information specified in 3.10.11.3.1 through 3.10.11.3.7 shall be included in the "Plan for Maintenance".

3.10.11.3.1 Periodic maintenance and logistic requirement. A program shall be established for all maintenance actions. To be identified are maintenance actions to be performed at three month intervals onboard ship by ship's force, and other actions and scheduling for these actions which shall require other facilities or personnel. This program shall include all actions required during the assumed 12 year equipment lifetime.

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3.10.11.3.1.1 The "Plan for Maintenance" shall summarize on a time schedule all parts which are to be replaced as defined by the preventive maintenance requirements, to provide reasonable confidence that the item will perform in accordance with the design operating life requirement until the next preventive maintenance action. Each part should be coded to indicate possible sources of supply for repeat buys or whether unrestricted competition is possible. Provisioning documentation, submitted in accordance with MIL-P-15137, shall reflect the "Plan for Maintenance" as approved. The "Plan" shall indicate whether an item of work should be conducted onboard the ship the equipment is installed or at another facility.

3.10.11.3.1.2 The following criteria shall be used in determining shipboard spare and repair part and equipment related consumables detailed in the "Plan for Maintenance":

(a) Insurance items (items which do not have a predicted usage).
  (1) Only those insurance items vital to the support of the primary mission of a ship or unit, or vital to the safety and welfare of personnel onboard ship shall be provided.
  (2) Insurance items shall be included in minimum depth (either unity or minimum replacement unit to meet potential 90 day combat requirements) in those cases where they are vital to ship primary missions.

(b) Whenever usage of a part during combat will increase from peace-time requirements, the combat requirements shall be noted as a technical over-ride. Further, when an extended period between expected replacements of a shipboard part will exist (exceeds approximately one year), its required continuation in the system shall be noted as a technical over-ride in the "Plan for Maintenance" and provisioning data, as submitted. (This notation is necessary to eliminate the possibility of removal of the item from the allowance list because of no replenishment usage within the supply system for two years.)

(c) The "Plan for Maintenance" shall reflect the military essentiality of each item.

3.10.11.3.2 Maintenance tasks. Maintenance tasks are hereby defined as procedural steps required to accomplish a preventive (including visual inspection) or corrective maintenance or calibration action. Contained in this section shall be a set of maintenance tasks for every maintenance action described in the "Plan for Maintenance". All necessary steps, unless self-evident to the person carrying out the task, shall be completely described. The frequency of occurrence of each task shall be clearly designated.

3.10.11.3.3 Installation and checkout procedures. Exact and complete procedures for installing, connecting and verifying all required inputs to the table to be carried out by the installing activity to insure operational integrity of the equipment. Each task shall contemplate all necessary steps, unless self-evident to the person making the installation. Special precautions such as possible erroneous connections, which could result in physical harm to workmen or damage to the equipment shall be provided in this section.

3.10.11.3.4 Personnel planning data. The "Plan for Maintenance" shall identify the personnel rating and skill levels required at each maintenance level. The data provided shall be complete and in sufficient detail to permit identification of acceptable personnel training programs and operational scheduling of personnel resources for both preventive and corrective maintenance.

3.10.11.3.5 Logistic documentation. Identification of all required documents (publications, engineering drawings, maintenance aids, etc.) necessary to accomplish each maintenance requirement and task.

3.10.11.3.6 Support equipment. All necessary special test equipment or special tools required to accomplish preventive or corrective maintenance or calibration of the equipment shall be detailed. The test equipment in this section shall be identified with the specific maintenance tasks for which it is required, and shall indicate possible alternate type of equipment which may be used. Specific NAVSEC (see 6.4) approval of the "Plan for Maintenance" shall be obtained before final equipment design may be considered approved. At any time that production tests indicate that the "Plan for Maintenance" requires revision, two copies of the proposed modification shall be forwarded to NAVSEC for approval.

3.10.11.3.7 Portions of the "Plan for Maintenance", as approved, shall be incorporated in the equipment technical manuals (MIL-M-24100) by the contractor as required by NAVSEC (see 6.4).

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3.11 Drawings. Drawings shall be as specified in the contract or order, except that the assembly drawings shall contain a warning to the effect that great care should be taken to maintain the specified tolerances during overhaul procedure and to point out the specially critical features required to be maintained for quietness. The location of all attachment points for pickups used in measuring structureborne noise shall be shown on the drawings, and suitable means shall be indicated for the protection and preservation of these attachment points.

3.12 Identification. The DRAI shall be prominently identified by affixing a label plate bearing the following inscription "Quiet Equipment - Use Great Care in Maintenance."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The supplier shall submit to NAVSEC (see 6.4) for approval the testing procedures to be followed during the preproduction inspection (see 4.3) and quality conformance inspection (see 4.4).

4.2 General inspection. The procedures for the various examination and tests to be performed on the equipment shall be as specified hereinafter. No adjustments of the DRAI calibration circuits shall be made after the start of preproduction and quality conformance inspection.

4.3 Preproduction inspection. Preproduction inspection shall consist of the examination and tests specified in table I. Tests shall be performed in the order listed.

Table I - Preproduction inspection

Examination and tests	Requirement reference	Test reference
General examination	MIL-I-983	4.5
Shielding and radio frequency noise reduction	MIL-I-983	MIL-I-983
Airborne noise	MIL-I-983	4.6.2
Structureborne noise	MIL-I-983	4.6.2
Synchronization	3.7.10	4.6.3
Effect of roll and pitch	3.5.12	4.6.4
Enclosure	MIL-I-983	MIL-I-983
Phase shift	3.7.8	4.6.5
Effect of magnetic fields	3.5.2	4.6.6
Transients	MIL-I-983	MIL-I-983
Accelerated life (includes tests for supply line voltage, accuracy, temperature and humidity, dielectric strength, insulation resistance, shock, vibration and inclination)	3.7.11 and MIL-I-983	4.6.1, 4.6.7, 4.6.8 and MIL-I-983

4.4 Quality conformance inspection.

4.4.1 Group A. All equipment offered for delivery on the contract or order shall be subjected to the group A examination and tests specified in table II. Failure to conform to this specification for any group A examination and test shall be counted as a defect and corrective action shall be taken by the supplier on future units and on any DRAI produced on the contract or order. This action shall be approved by NAVSEC (see 6.4). In addition to the corrective action, the occurrence of a failure shall necessitate rerunning all of the group A tests and tests from group C as specified by NAVSEC.

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Table II - Group A examination and tests.

Examination and tests	Requirement reference	Test reference
General examination	MIL-I-983	4.5
Dielectric strength[1/]	MIL-I-983	MIL-I-983
Insulation resistance[1/]	MIL-I-983	MIL-I-983
Accuracy[2/]	3.7.11	4.6.1
Supply line voltage	MIL-I-983	MIL-I-983
Synchronization	3.7.10	4.6.3
Burn-in	3.10	4.6.9

[1/] For all but rotating components of equipment.
[2/] See 4.6.1.5.

4.4.2 Group C. Group C tests will be required by NAVSEC (see 6.4) when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be subjected to the group C tests specified in table III. In the event of failure to conform to this specification for any group C test, the supplier shall correct the cause of failure on future production units and repair the deficiency in any DRAI produced on the contract or order.

Table III - Group C tests.

Tests	Requirement reference	Test reference
Transients	MIL-I-983	MIL-I-983
Accelerated life	MIL-I-983	4.6.7
Enclosure	MIL-I-983	MIL-I-983
Vibration	MIL-I-983	MIL-I-983
Shock	MIL-I-983	MIL-I-983
Airborne noise	MIL-I-983	4.6.2
Structureborne noise	MIL-I-983	4.6.2

4.5 General examination. The general examination shall be conducted in accordance with MIL-I-983. The examination shall include, but shall not be limited to, the following checks:

(a) Counter slew check.
(b) Own ship's position remote slew check.
(c) Latitude and longitude synchro output check.
(d) Own ship's distance synchro output check.
(e) Relative to true bearing conversion check (CDX outputs).
(f) Manual speed check.
(g) Manual course check.
(h) Calibration controls check.
(i) Correct fuse value check.

4.6. Test procedures.

4.6.1 Accuracy.

4.6.1.1 Resultant error. The resultant error test shall be conducted concurrently with the test for accuracy of position. Both tests shall be conducted using the runs described in 4.6.1.2. Resultant errors shall be determined as indicated in 3.7.9.1 for each run.

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4.6.1.1.1 Transmitted distance signal accuracy test. In addition to the test specified in 4.6.1.1 the accuracy of the N-S and E-W distance signals shall be verified by the following test: With a constant speed input of 10 knots and an initial course of zero degrees (North) an octagon test course shall be run. The octagon test course shall be comprised of 45 degrees course changes and each side of the octagon shall be a course run of 30 minutes. The distance signal shall be considered as meeting this requirement if each side of the octagon and the comparison of the final position to the initial position are within the accuracy requirements (3.7.11).

4.6.1.2 Position error. The test for accuracy of position shall include the seven runs of 100 nautical miles each and one run of 1000 nautical miles shown in table IV (see Appendix for definition of abbreviations and method of computation). Runs for accuracy of position test will be conducted at input speeds as scheduled in table V in order to determine the accuracy of speed components. Accuracy check tests required by MIL-I-983 shall consist of runs "a", "b" and "g" only of table IV. The accuracy of position shall be determined in accordance with 3.7.9.2.

Table IV - Accuracy test runs.

Run	D	L_1	Course
		Degrees	True Azimuth
a	100	0	005
b	100	15S	095
c	100	25N	240
d	100	40S	300
e	100	55N	045
f	100	65S	270
g	100	85N	090
h 1/	500	40N	045
i	1000	40N	045

1/ Data for run "h" shall be obtained when practicable by making a stop for readings at 500 nautical miles during run "i".

4.6.1.3 Accuracy of speed components. Runs "a", "b", "c" and "d" shall be conducted at various input speeds as indicated in table V. After each change of speed, readings shall be taken and used to determine the amount of error in output components of own ship's speed resulting with the input speeds listed in table V. The accuracy of speed component outputs shall be determined in accordance with 3.7.9.3.

Table V - Input speed during accuracy tests.

Hour	1st	2nd	3rd	4th	5th	6th	7th
Run	Speed in knots						
a and c	40	2	5	3	30	20	---
b and d	20	4	10	6	15	25	20
e thru i	Any selected constant speed between 10 and 30						

4.6.1.4 Integrator error. The integrator error test shall be conducted at speeds of 4 and 40 knots. Integrator errors shall be determined as indicated in 3.7.9.4 for each run. Run time for each test shall be of sufficient length to demonstrate integrator accuracy.

4.6.1.5 Accuracy (group A inspection). When conducting quality conformance inspection, group A (see table II), the accuracy tests required may be reduced to runs "a", "b" and "g" only, if the results of the reduced tests meet the requirements of 3.7.11.

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4.6.1.6 The accuracy tests of 4.6.1 through 4.6.1.4 shall be performed after each of the examinations and tests listed in tables I through III. In addition, the accuracy tests of 4.6.1 through 4.6.1.4 shall be performed during the following tests: the effect of roll and pitch, enclosure, the effect of magnetic fields, and the accelerated life test. (The supply line voltage, temperature and humidity, and vibration and inclination tests shall be the only phases of the accelerated life test during which accuracy testing is performed.)

4.6.1.7 The general examination, shielding and radio frequency noise reduction, airborne and structureborne noise, synchronization, dielectric strength and insulation resistance phases of the accelerated life test may be excepted from the tests of 4.6.1.6.

4.6.1.8 The results of all of the required accuracy tests shall meet the requirements of 3.7.11. The abbreviated accuracy tests of 4.6.1.5 may be performed whenever the accuracy tests must be performed during the tests listed in 4.6.1.6.

4.6.1.9 Speed counter errors. The accuracy of the speed counters shall be measured with respect to the transmitted or received signals (see 3.7.6 and 3.7.11). Readings shall be taken at speeds of 1 knot, 10 knots, zero knots and 40 knots in that order.

4.6.2 Airborne and structureborne noise. -

4.6.2.1 Airborne noise. Airborne noise levels shall not exceed the 1/3 octave band sound power level limits for equipment as shown in table VI.

Table VI - One-third octave band limits for airborne noise

(sound power level re 10^{-13} watts).

Band center frequency (cps)	Noise level db.	Band center frequency (cps)	Noise level db.
25	1/	630	72
31.5	1/	800	70
40	92	1000	68.5
50	90	1250	67
63	88.5	1600	65
80	87	2000	63.5
100	85	2500	62
125	83.5	3150	60
160	82	4000	58.5
200	80	5000	57
250	78.5	6300	55
315	77	8000	53.5
400	75	10000	52
500	73.5		

1/No limits established.

4.6.2.2 Structureborne noise. Structureborne noise levels shall not exceed the values shown on figure 2 for equipment in any 1/3 octave band for the frequencies specified in 4.6.2.4. If 1/3 octave band levels exceed the specified limits in any band, the equipment may be accepted on the basis of narrow band analysis showing no peaks over the limits specified for the 1/3 octave band or bands, in question. No unnecessary weight shall be added to the equipment under test.

4.6.2.3 One-third octave analysis. The levels of each 1/3 octave band for the selected pickup shall be recorded on a graphic level recorder. The complete frequency range shall be scanned twice. If any 1/3 octave band indicates a difference in recorded levels of more than 3 db. between the two scans, the level in this 1/3 octave band shall be recorded for one minute. The highest level recorded during this period shall be reported as the level for that band.

4.6.2.4 Frequency range. The frequency range shall be as specified in MIL-STD-740, except that if the lowest exciting frequency of the equipment under test is below the range specified in MIL-STD-740 the lower limit of frequency shall be the lowest exciting frequency of the equipment. Frequencies referred to herein are the band center frequencies.

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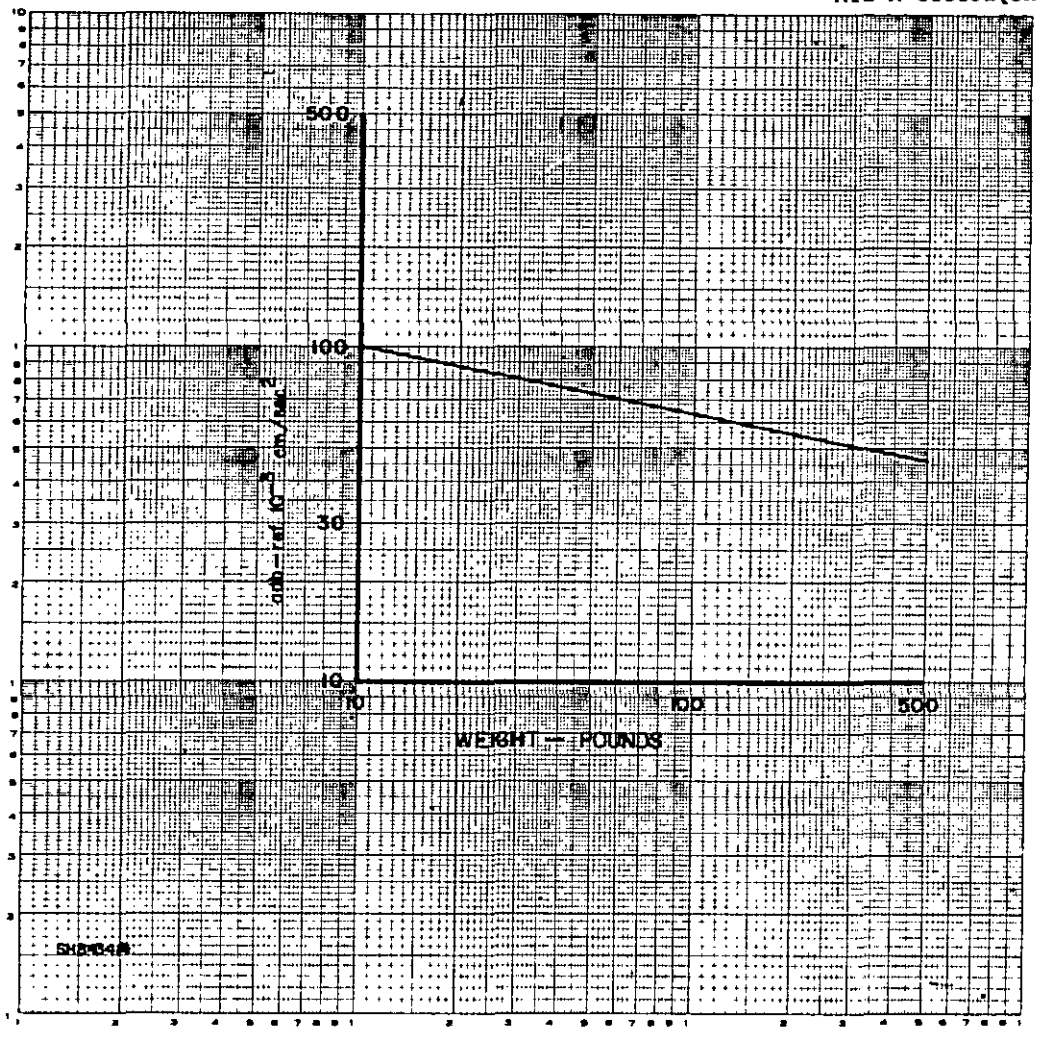


Figure 2 - Acceptance limits for structureborne noise.

4.6.2.5 Mounting for test. The equipment shall be mounted on a fixture by the designed points of attachment of the equipment. This fixture shall be stiff between points of attachment and damped to eliminate resonance within itself. If the equipment being evaluated contains internal sound isolating mounts the mass of the fixture shall be great enough to permit these internal mounts to function properly. (In order to do this, the mass of the nonisolated structure, including the added fixture should be approximately ten times the mass of the internal structure which is supported on sound-isolation mounts.) The combined assembly of equipment and test fixture shall be resiliently mounted and oriented in its normal operating position. The resilient mounting may consist of mounts as specified in MIL-STD-740 or a suspension system using flexible cord (shock cord). Preference shall be given to the use of resilient mounts. The total weight on the resilient mounting system (including the added fixture) shall be used to determine the allowable limits from figure 2. Equipment resilient mounting shall not be used during this test.

4.6.2.6 Attachment of pickups. Pickups shall be attached directly to the equipment being tested without the use of mounting blocks, except in those instances where there is insufficient space to accommodate the tri-axial array of pickups. Where mounting blocks are necessary in order to accommodate the tri-axial array of pickups, the blocks shall be welded or brazed to the equipment, and the block surfaces shall be accurately machined. The pickups may be attached to the added fixture, if necessary, but they should be close to the equipment's own attachment points.

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4.6.2.7 Operating conditions. All external electrical design loads shall be connected to the DRAI during low noise testing. Speed and course inputs shall be continuously slewed at the following rates and amplitudes:

Course - Plus 350 degrees at 10 degrees + 2 degrees per second.
Speed - One to 30 knots cyclically at 1 knot per second.

4.6.3 Synchronization. Tests shall be made to ensure that the DRAI will be self-synchronous as specified in 3.7.10.

4.6.4 Effect of roll and pitch.

4.6.4.1 The purpose of this test shall be to determine whether the DRAI will operate satisfactorily under condition of roll and pitch.

4.6.4.2 The unit under test shall be mounted on a Scorsby test stand which shall be operated during the runs with an amplitude of roll of 40 degrees on each side of the vertical with a period between 8 and 10 seconds, and an amplitude of pitch of 10 degrees on each side of the vertical with a period between 6 and 8 seconds. The unit may be mounted in any one or more positions in which it may be required to operate abroad and the roll and pitch axes of the Scorsby test stand may be interchanged, if necessary, to provide clearance for the motion of the unit.

4.6.5 Phase shift. Tests shall be made to ensure that the phase shift caused by external synchro connected in the input circuit will not be detrimental to the operation of the DRAI as specified in 3.7.8.

4.6.6 Effect of magnetic fields. The accuracy tests of 4.6.1 shall be made to ensure that a magnetic field will not affect the accuracy of the DRAI as specified in 3.5.2.

4.6.7 Accelerated life. The accelerated life test shall be conducted as specified in MIL-I-983 except that the ambient temperature during hours 175-300 shall be 149°F., (65°C.) in lieu of the 131°F. (55°C.) specified. The accelerated life test shall be conducted with a continuous input of constantly varying synchro input of own ship's course at a rate of 360 degrees per minute reversing direction of rotation at least once every half hour and a constantly varying synchro input of speed of simple harmonic motion of from 5 to 35 knots with a 60-second period for 500 hours. The heat dissipated from the sides and back of the equipment shall be restricted by means of low heat conducting bulkheads placed adjacent to the sides and back of the equipment.

4.6.8 Shock and vibration. The DRAI shall conform to the grade A, class II, H.I. (High Impact) shock requirements of MIL-S-901. Equipment shall meet the vibration test for vital equipment of MIL-I-983.

4.6.9 Burn-in tests. Upon completion of the "Group A" tests (see 4.4.1) all production DRAIs shall be subjected to a burn-in test. The first production DRAI shall be subjected to a 1000 hour test and all following DRAIs shall be subjected to a 100 hour test as specified below. No calibration, alignment, adjustment, modification, or replacement of any module, subassembly, component or part is permitted during the burn-in tests. All burn-in tests shall be conducted on the DRAI in the condition that it completes the "Group A" tests. The DRAI shall be considered as having failed the burn-in test if the DRAI fails to meet any of the accuracy requirements or; if any part, component, subassembly or module requires replacement, adjustment or calibration or; if the DRAI fails to operate continuously during each 250 or 25 hour phase. The only exception to this failure definition shall be a failure of the light bulbs used to illuminate the controls and readouts on the DRAI front panel. Failure of these bulbs shall not be considered a DRAI burn-in test failure. NAVSEC (see 6.4) shall be notified of all failures and corrective action approval acquired prior to correction incorporation or continuation of test. Submission of proposed corrective action for approval shall include the following:

(a) A description of the failure.
(b) A discussion of the cause of the failure.
(c) A description of the proposed corrective action.
(d) An analysis of how the reliability and maintainability requirements (see 3.10) are effected by both the failure and the corrective action.

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Both the 1000 and 100 hour tests shall consist of four phases as follows:

(a) Phase one - 149°F.
(b) Phase two - Room temperature.
(c) Phase three - 149°F.
(d) Phase four - Room temperature.

Each testing phase shall consist of the DRAI being continuously operated at the testing phase temperature for the prescribed time and then a complete accuracy test at room temperature (see 4.6.1). Each phase of the 1000 hour test shall be of 250 hour duration and of 25 hour duration for the 100 hour test. The DRAI shall experience a significant speed and course change every hour and each phase shall be conducted at a different latitude. Inputs shall be so selected as to represent the full DRAI operational range. A failure prior to the completion of phase three shall require correction and the entire burn-in test started over again. A failure in phase four shall require correction and a retest consisting of phases three and four. A failure in the retest shall require correction and another retest consisting of phases three and four. In the event that any three failures have occurred, NAVSEC (see 6.4), shall be so notified, a correction report submitted, a design evaluation conference held and NAVSEC (see 6.4) shall decide what preproduction tests (see 4.3) shall be conducted on the redesigned DRAI. The corrective action shall apply to all DRAIs on the contract or order. The second and third failure need not be a duplicate of the first failure to be considered a failure. Upon completion of the above test the DRAI shall be refurbished as required by the reliability and maintainability requirements (see 3.10), accuracy tested and shipped.

4.7. Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Dead reckoning analyzer-indicator.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation, and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination.

5.1.2 Onboard repair parts. Onboard repair parts shall be preserved and packaged by level A; packed by level C, and marked by levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation and packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.2).

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(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging, packing and marking. The equipment and accessories, repair parts and publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified, and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for Naval service where it is expected to withstand continuous use for long periods, under Military service conditions without benefit of overhaul. The equipment is in each case a vital instrument intended for important use by the forces concerned. Failure at a critical moment invariably results in serious reduction in the battle efficiency of the ship. Functional use is indicated in 3.3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Inventory control point for repair parts (see MIL-I-983).
(d) Whether FSN is to be entered on the identification plate (see MIL-I-983).
(e) Level of preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.3 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 NAVSEC. The applicable branch in NAVSEC is the Interior Communication, Navigation, Control and Computer Systems Branch.

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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APPENDIX

COMPUTATIONS

10. Definition of abbreviations. The abbreviations used in describing the methods of computing positions are defined as follows:

D - Distance of run in nautical miles.
L_1 - Initial latitude setting.
L_2 - Calculated final latitude.
L_M - Estimated middle latitude.
Course angle - Direction of movement measured in degrees in direction from true north and south.
C - Course setting - Direction of movement measured in degrees clockwise from true north.
DL - Difference in latitude between initial setting and final counter reading.
DL_o - Difference in longitude between initial setting and final counter reading.
K - The value of nautical miles per minute of longitude at the final latitude.
D_E - Distance East.
D_N - Distance North.
Ep - Position error.

20. The Mercator method of computing the calculated final position for all runs except when the course is 90 degrees or 270 degrees is shown in the following example (calculations for run "d"):

(a) Given initial latitude = 40°00.0'S = L_1
initial longitude = 0°00.0'E
course = 300° = C
distance = 100 miles = D

(b) D_E = 100 sin 60° = 86.603 miles
D_N = 100 cos 60° = 50.000 miles
L_M = L_1 + 1/2 D_N
= 40°00.0'S + 25.0' = 40°25.0'S (1 mile = 1 minute of latitude)

(c) The difference in latitude is:
$DL = 100 \cos 60° \times \frac{60.000}{59.949} = 50.043$ minutes
in which 59.949 is the value of miles per degree of latitude at the estimated middle latitude (L_M), from table 6 in Bowditch (HO Publication No. 9)

(d) The final latitude is:
$L_2 = L_1 + DL$ = 40°00.0' - 50.043' = 39°09.557'S

(e) From explanation of table 5 in Bowditch, the meridianal parts for the initial latitude is:
m_1 - 2607.7

(f) Similarly the meridianal parts for the final latitude is:
m_2 - 2542.5

(g) The difference in longitude is:
$DL_o = (m_2 - m_1) \tan C$
= 65.2 tan 60°
= 112.92' = 1°52.92'

30. When the course is 90 degrees or 270 degrees, the final position is computed by the Middle Latitude sailing method, as shown in the following example:

(a) Given L_1 = 65°00.0'S
C = 270°
D = 100 miles

(b) The difference in latitude is zero.

(c) The difference in longitude is:
$DL_o = 100 \sin 270° \times \frac{60.000}{25.474} = 3°55.53'$
in which 25.474 is the value of miles per degree of longitude corresponding to 60° latitude (the mean latitude during the run)(from table 6 in Bowditch).

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40. The position error is computed as shown in the following example (calculated for run "d"):

(a) Given: Observed values:
DL 0° 50.7'
DL_o 1° 50.5'

From table IV herein:
DL 0° 50.04'
DL_o 1° 52.06'
K .778

(b) The latitude error is:
0° 50.7' - 0° 50.0' = 0.7'
Since the errors are small as compared with the total distance, 1 minute of latitude may be considered as equal to 1 nautical mile. Then

North component error = 0.7 mile.

(c) The longitude error is:
1° 50.5' - 1° 52.1' = 1.6'
The value K of miles per minute of longitude may be found by dividing the value of miles per degree at the final latitude (from table 6 in Bowditch) by 60.
$K = \frac{46.7}{60.0} = 0.778$
The east component error is K times the longitude error in minutes, or 1.24 miles.

(d) The position error is the vector sum of the north and east component errors, expressed in percent of the distance, or

$$E_p = \sqrt{(1.24)^2 + (0.7)^2} \times \frac{100}{100}$$

$$= \sqrt{2.03} \times \frac{100}{100} = 1.42 \text{ percent}$$

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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